New Jersey
Student Learning
Assessments-Adaptive

Practice Test

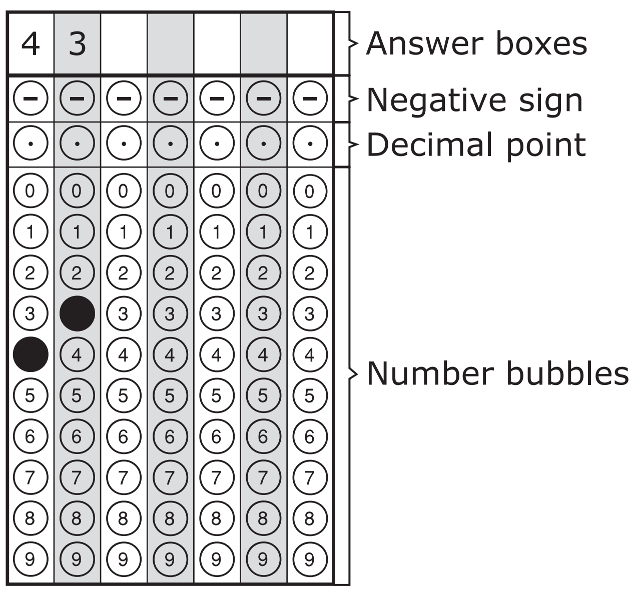
Algebra I

Student Name ______________________________

Directions:

Today, you will take the Algebra I Practice Test. You will be able to use a calculator. If you do not know the answer to a question, you may go on to the next question. When you finish, you may review your answers and any questions you did not answer.

1. For questions with bubbled responses, fill in the circle next to your answer choice. If you change your answer, make sure you erase your old answer completely. Do not cross out or make any marks on the other choices.
2. For questions with response boxes, write your answer neatly, clearly, and in the space provided.
3. For questions with response grids, write your answer in the answer boxes at the top of the grid.
   - Write only one digit or symbol in each answer box.
   - Use only the options listed below each box.
   - Be sure to write a negative sign or decimal point in the answer box if it is part of the answer.



1. The gravitational potential energy of an object is given by the formula $P = mgh$.

Which equation is correctly solved for the height, h?

Ⓐ $h = P + mg$

Ⓑ $h = P - mg$

Ⓒ $h = \frac{P}{mg}$

Ⓓ $h = Pmg$

2. The function $h(p)$ represents the cost for a company to manufacture p posters.

What is the domain of the function?

Ⓐ all integers

Ⓑ all real numbers

Ⓒ all non-negative integers

Ⓓ all positive rational numbers

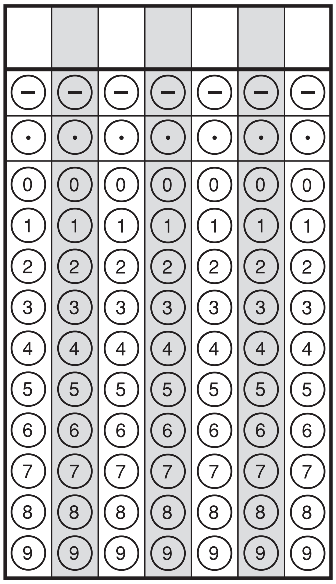
3. A system of equations is shown.

$$4c + 2d = 11$$

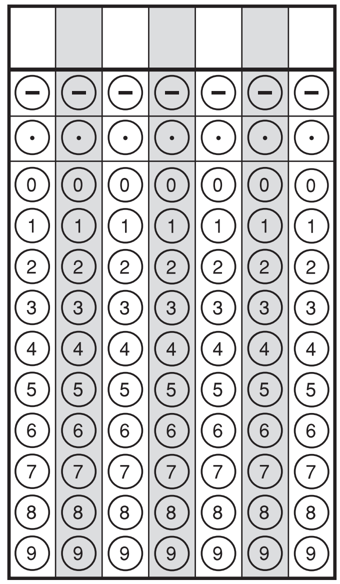
$$\frac{7}{2}d = 41 - 22c$$

What is the solution to the system?

$c =$

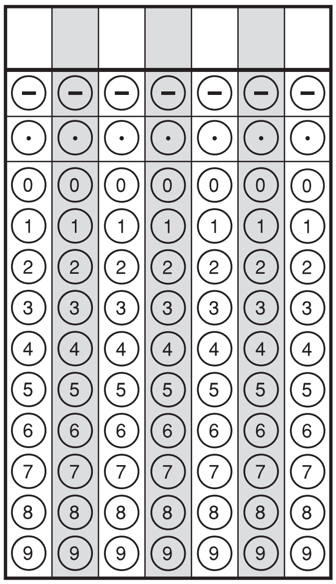


$d =$



4. The function $f(t) = -16t^2 + 20t + 4$ gives the height of a ball, in feet, t seconds after it is tossed.

What is the average rate of change, in feet per second, over the interval [0.75, 1.25]?



5. Select all of the values of a correlation coefficient that suggest a strong linear relationship between two variables.

Ⓐ 0.8

Ⓑ 0.4

Ⓒ 0

Ⓓ –0.1

Ⓔ –0.9

GO ON ▶

6. A quadratic equation is shown.

$$0 = x^2 - 3x - 4$$

Which value is a solution to this equation?

Ⓐ 1

Ⓑ 2

Ⓒ 3

Ⓓ 4

7. In 2015, Macon County had a population of 53,792. The population increases by 2.5% annually.

Which function can be used to model the population t years after 2015?

Ⓐ $f(t) = 1.025t + 53{,}792$

Ⓑ $f(t) = 1.25t + 53{,}792$

Ⓒ $f(t) = 53{,}792(1.025)^t$

Ⓓ $f(t) = 53{,}792(1.25)^t$

8. A rectangular park has an area of 250 square feet. The length of the park is 7 feet more than twice the width, w, of the park.

Create an equation in terms of w to model this situation.

9. A function is shown.

$$h(t) = -t^2 + 10t - 16$$

For which interval of t-values is the function both positive and increasing?

Ⓐ $t < 5$

Ⓑ $t > 8$

Ⓒ $2 < t < 5$

Ⓓ $5 < t < 8$

10. The number of visitors, in millions, to Bryce Canyon National Park from 2010 to 2014 can be modeled by the expression $1.281(1.024)^t$, where t is the number of years since 2010.

What is the factor, after t years, by which the number of visitors has grown since 2010?

Ⓐ 1.281

Ⓑ 1.024

Ⓒ $(1.024)^t$

Ⓓ $1.281(1.024)^t$

GO ON ▶

11. A function is shown, where b is a real number.

$$f(x) = x^2 + bx + 144$$

The minimum value of the function is 80.

Create an equation for an equivalent function in the form $f(x) = (x - h)^2 + k$.

$f(x) =$ $\boxed{}$

12. Juan records the number of hot dogs sold and the number of cars that drive by a hot dog stand each hour. His data are shown in the table.

Juan's Data

Hour	Number of Hot Dogs Sold	Number of Cars
1	30	73
2	9	32
3	21	56
4	0	11
5	24	62

Based on his data, which conclusion can Juan make?

Ⓐ An increase in cars is associated with a decrease in hot dog sales.

Ⓑ An increase in cars is associated with an increase in hot dog sales.

Ⓒ An increase in cars causes a decrease in hot dog sales.

Ⓓ An increase in cars causes an increase in hot dog sales.

GO ON ▶

13. This item has two parts.

Some friends spent a total of \$12.00 on popcorn and drinks at the movie theater. A bucket of popcorn cost \$2.00 and a drink cost \$1.50.

Part A.

Create an equation to represent the relationship between the number of buckets of popcorn, x, and the number of drinks, y, the friends bought for \$12.00.

Part B.

The friends bought 4 drinks.

How many buckets of popcorn did they buy?

GO ON ▶

14. The model $n(t) = 2^t$ represents the number of bacteria in a petri dish after t hours, where $t = 0$ represents the time when the bacteria were first put into the dish.

What is the correct value and interpretation of $n(8)$?

Ⓐ $n(8) = 256$, so after 8 hours there are 256 bacteria.

Ⓑ $n(8) = 256$, so after 256 hours there are 8 bacteria.

Ⓒ $n(8) = 3$, so after 8 hours there are 3 bacteria.

Ⓓ $n(8) = 3$, so after 3 hours there are 8 bacteria.

15. Joshua draws two squares. The area of square M is $\frac{3}{4}$ the area of square N.

Create an equation that relates the side length of square M, s, and the area of square N, A.

STOP

Paper Practice Test Answer and Alignment Document: Math

Algebra I

Item Sequence	Answer Key	Standards Alignment
1	C	A.CED.A.4
2	C	F.IF.B.5
3	1.45 and 2.6, or any equivalent values	A.REI.C.6
4	–12, or any equivalent values	F.IF.B.6
5	A, E	S.ID.C.8
6	D	A.REI.B.4b
7	C	F.LE.A.2
8	$2w^2 + 7w = 250$, or any equivalent equation	A.CED.A.1
9	C	F.IF.B.4
10	C	A.SSE.A.1a
11	$(x-8)^2 + 80$ or $(x+8)^2 + 80$, or any equivalent expressions	F.IF.C.8a
12	B	S.ID.C.9
13	Part A: $2x + 1.5y = 12$, or any equivalent equation Part B: 3, or any equivalent value	A.CED.A.2
14	A	F.IF.A.2
15	$s^2 = \frac{3}{4}A$, or any equivalent equation	A.CED.A.2